

N	пчм	масса, г.
1		0,511
2		0,512
3		0,506
4		0,509
5		0,510
6		0,499
7		0,510
8		0,510
9		0,506
10		0,505

$$M = 2905 \pm 5$$

$$L = 2,68 \text{ м}$$

$$J = 1,7$$

$$T_1 = (70,36 - 4,17) \cdot \frac{1}{10}$$

$$T_2 = (50,6 - 0,64) \cdot \frac{1}{10}$$

$$M_{\text{сред.1}} = 713,42 \pm 0,12$$

$$M_{\text{сред.2}} = 713,52 \pm 0,12$$

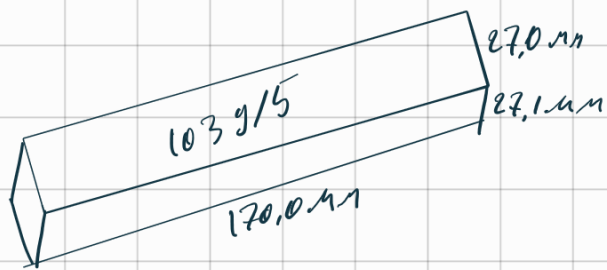
$$R \sim 0,33 \text{ м}$$

$$V = 0,122 \text{ м}$$

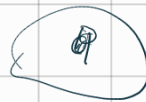
N	X
1	5,4
2	5,6
3	5,6
4	5,7
5	5,7

N	ΔX
6	$\sim 0,012$
7	$\sim 0,012$
8	$\sim 0,012$
9	$\sim 0,012$
10	$\sim 0,012$

на бугер



$(721/1) \rightarrow$
 $(721/5) \rightarrow$



m, g

h, mm

V, mm^3

720,7

22,1

90,0

721,1

22,2

90,0

1039,2

771,7

881,2

для вытеснения:

N	T
1	0.04.45
10	0.43.95
20	1.27.00
40	2.55.65

для кольца:

N	T
1	0.04.14
5	0.20.80
10	0.41.75
20	1.23.48
30	2.05.24

для гирь:

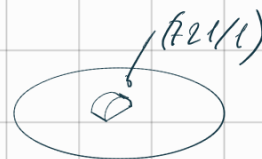
N	T
1	0.04.20
5	0.19.06
10	0.37.69
20	1.14.74
30	1.51.91

кольцо + диск

N	T
1	0.03.74
5	0.18.79
10	0.37.65
20	1.15.26
30	1.52.84

для гири (ушки без)

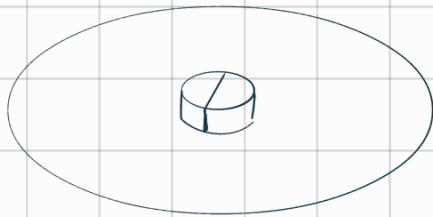
1	0.03.58
5	0.17.81
10	0.35.60
20	1.11.17
30	1.46.72



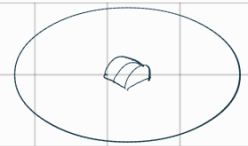
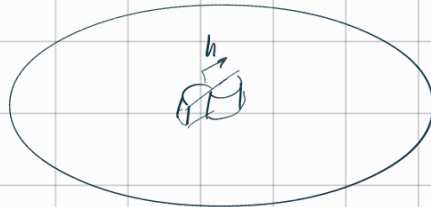
1	0.03.41
5	0.17.24
10	0.34.45
20	1.08.81
30	1.43.21



1	0.03.44
5	0.17.23
10	0.34.45
20	1.08.86
30	1.43.25



1 0.03.07
 5 0.15.35
 10 0.30.76
 20 1.01.48
 30 1.32.21



1 0.03.01
 5 0.15.06
 10 0.30.12
 20 1.00.27
 30 1.30.39

h, m	T	$5T$	$10T$	$20T$	$30T$
0.5	3:08	15:45	30:89	1.01:65	1.32.51
1	3:09	15:46	30:95	1.01:94	1.32.92
1.5	3:13	15:63	31:22	1.02:48	1.33:71
2.	3:18	15:89	31:72	1.03:43	1.35.11
2.5	3:23	16:17	32:26	1.04:50	1.36.72
3	3:29	16:39	32:84	1.05:63	1.38.42
3.5	3:36	16:75	33:48	1.06:87	1.40.88
4	3:44	17:20	34:38	1.08:74	1.43.12
4.5	3:53	17:61	35:21	1.10:38	1.45.54
5	3:66	18:18	36:32	1.12.52	1.48.82
5.5	3:76	18:71	37:37	1.14.73	1.52.08
6	3:86	19:25	38:42	1.18:82	1.55.21
6.5	3:92	19:83	39:59	1.19:13	1.58:69
7	4:14	20:45	40:89	1.21:75	2.02.60
7.5	4:22	21.09	42:16	1.24.29	2.06.41
8	4:35	21.77	43:53	1.27.02	2.10.50

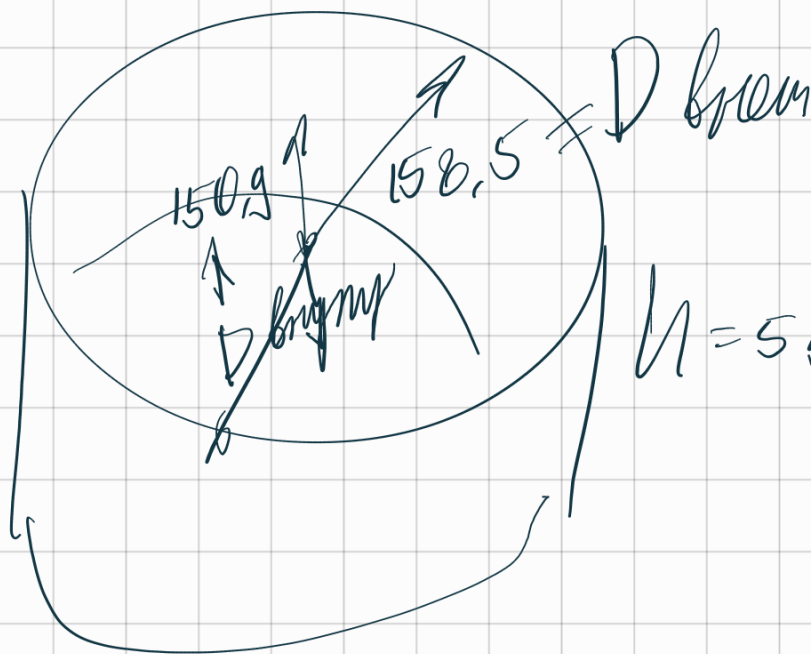
$L = 2140 \text{ mm}$ - гребень шланга

$$D = 250,1 \pm 0,1$$

$$h = 30,5 \pm 0,3$$

$$\frac{2140}{2140} \quad 0,2 \quad \frac{0,2}{2140}$$

$$\frac{2140^2}{2140} \quad 0,4 \quad \frac{0,4}{2140}$$

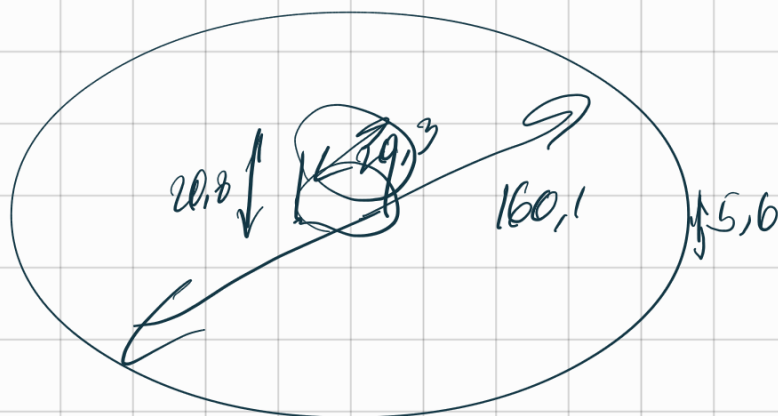


$$h = 55,7$$

Δx

$$0,4 \cdot 2140 + 0,2 \cdot 125$$

$$\frac{1}{2} \frac{\Delta x}{x}$$



$$I = I_0 + mR^2$$

1585

$$I = k \cdot h^2 + b$$

$$I_0 = (101378 \pm 1) \cdot 10^{-8}$$

$$m_0 = (1,585 \pm 0,008) \mu\text{m}$$

$$0,106138$$

$$0,009661$$

$$0,005$$

$$0,008$$

масса $\pm 0,12$

Стружка 1637,0

① Призма 1 46,2

②+ Призма 2 58,5

ГРУЗ 1 2456,2

ГРУЗ 2 2554,2

⑥



$N = 500$
 $T = 11:41:46$
 $T_1 =$

Масса магнитов 6752,1 г.

Длина стружки 100,06 см

Разнообразие $L = 50$ см

$$L_1/L_2 = 3$$

ТЕОР. РАСЧЕТ

④ $b_2 = 8$ см
 $b_1 = 24,9$ см

⑨

$N = 100$

$T = 141,47$

$T_2 = 1,4147$

⑦



$N = 500$
 $T = 11:47:20$
 $T_2 =$

Реальность:

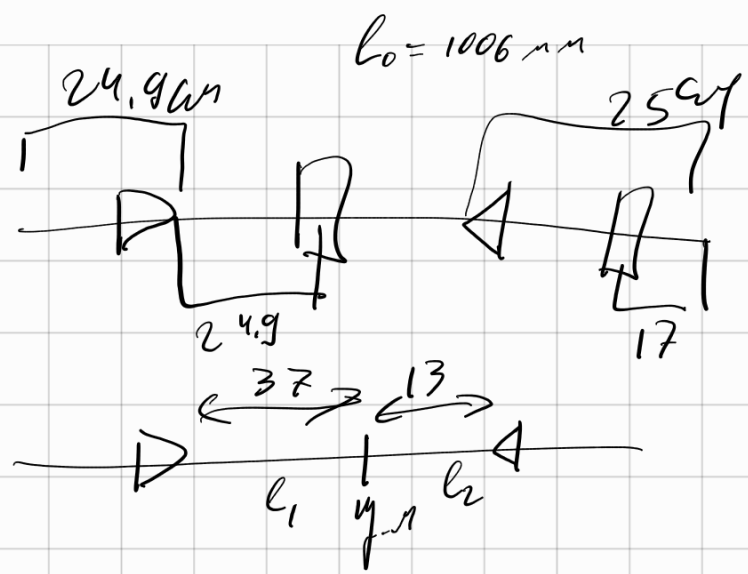
③ $L =$

⑧

$$\frac{T_2 - T_1}{T_1} \approx 0,7\%$$

5) $l_1 =$
 $l_2 =$

10)



11)

$$g = (2\pi)^2 \frac{L}{T^2} (1 + 2\beta E) =$$

$$\beta = \frac{l_2}{l_1 - l_2} = 10,1 \frac{\text{m}}{\text{cm}}$$

$$E = \frac{T_2 - T_1}{T_1}$$



$5^\circ N = 300$

$7,5^\circ N = 1$
 2%



$5^\circ N = 500$

$7,5^\circ N_1 = 1$

$$(2\pi)^2 \cdot \frac{L}{T^2} \left(1 + 2 \frac{(T_2 - T_1)}{T_2} \frac{l_2}{(l_1 - l_2)} \right)$$

3%?

10%

$$T_2 - T_1 = 11 \pm 6 \text{ nC.}$$

$$\delta(T_2 - T_1) = \frac{6}{11}$$

$$T_2 = 1414 \pm 3$$

$$\delta(T_2) = \frac{3}{1414}$$

$$\delta(l_2) = \frac{1}{130}$$

$$\delta(l_1 - l_2) = \frac{2}{240} = \frac{1}{120}$$

$$\Delta = 0,0047498268$$

$$10,2 \pm 0,2$$

$$10,3 \pm 0,4$$

$$10,3 \pm 0,2$$

$$10,3 \pm 0,3$$

$$10,3 \pm 0,3$$

$$10,3 \pm 0,2$$

$$10,3 \pm 0,2$$

$$10,3 \pm 0,3$$

$$10,2 \pm 0,3$$

$$10,2 \pm 0,3$$

$$10,27 \pm 2,7$$

$$10,3 \pm 0,3$$

$$\delta T = \delta T = \frac{T \cdot \Delta T}{T}$$

V_{Z2}	T_{Z2}	TIME/DIV	T_c	V_{Z2}	ΔV_{Z2}	$\delta V_{Z2}, T_{Z2}$
1,000 2 KHz	25	$0,2 \cdot 10^{-3}$	0,001	1000	40	0,4
50,76 Hz	20	$5 \cdot 10^{-3}$	0,02	50	2,5	0,75
500,17 Hz	20	$0,5 \cdot 10^{-3}$	0,002	500	25	0,17
1,0123 MHz	25	$0,2 \cdot 10^{-6}$	10^{-6}	10^6	$40 \cdot 10^3$	12300
20,070 KHz	25	$10 \cdot 10^{-6}$	$50 \cdot 10^{-6}$	$20 \cdot 10^3$	800	70
200,22 KHz	25	10^{-6}	$5 \cdot 10^{-6}$	$200 \cdot 10^3$	8000	220

3) $U_{max} = \frac{4 \cdot 5}{2} = 10 \text{ V}$ $\Delta U_{max} = 5/5 = 1 \text{ V}$ $\delta U_{max} = 0,1$

$U_{min-40dB} = \frac{2,4}{2} \cdot 5 \cdot 10^{-3} = 6 \text{ mV}$ $\Delta U_{min-40dB} = \frac{5 \cdot 10^{-3}}{5} = 1 \text{ mV}$ $\delta U_{min-40dB} = \frac{1}{6}$

$U_{min-20dB} = \frac{5,8}{2} \cdot 20 \cdot 10^{-3} = 58 \text{ mV}$ $\Delta U_{min-20dB} = \frac{20 \cdot 10^{-3}}{5} = 4 \text{ mV}$ $\delta U_{min-20dB} = \frac{4}{58}$

$U_{min} = \frac{2,4}{2} \cdot 0,5 = 0,6 \text{ V}$ $\Delta U_{min} = \frac{0,5}{5} = 0,1 \text{ V}$ $\delta U_{min} = \frac{1}{6}$

$$20 \lg \frac{U_{max}}{U_{min-20dB}} = 44,7 \text{ dB}$$

$$20 \lg \frac{U_{min-20}}{U_{min-40dB}} = 19,7 \text{ dB}$$

$$20 \lg \frac{U_{min-40dB}}{U_{min}} = -40 \text{ dB}$$

5)

$$U_0 = 6 \text{ B}$$

$$U_{max} = 5,258 \text{ MHz}$$

$$U_{max} = 5,2 \text{ B}$$

$$K = \frac{13}{15} = 0,87$$

ν	U	K
10^2	6	1
10^4	6	1
10^5	6	1

СТА116

$$L = 600,0 \pm 0,5 \mu\text{m}$$

Δ10РА116

$$f_1 \quad 4,12 \ 548$$

$$f_2 \quad 8,26 \ 222$$

$$f_3 \quad 12,38 \ 14$$

$$f_4 \quad 16,51 \ 20$$

$$f_5 \quad 20,63 \ 62$$

$$f_1 \quad 4,23 \ 598^{30}$$

$$f_2 \quad 8,47 \ 632$$

$$f_3 \quad 12,70 \ 98$$

$$f_4 \quad 16,94 \ 51$$

$$f_5 \quad 21,17 \ 15$$

μΔ6

$$f_1 \quad 3,27 \ 732 \quad \kappa \Gamma_y$$

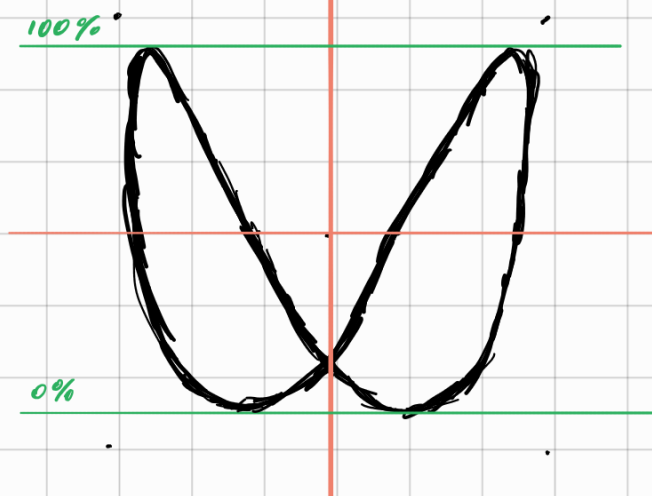
$$f_2 \quad 6,50 \ 957$$

$$f_3 \quad 9,74 \ 726$$

$$f_4 \quad 12,98 \ 74$$

$$f_5 \quad 16,22 \ 26$$

$$\frac{f_1}{2}$$



Δ05Ρ0Т110СТ6

$$5 \quad f_0 = 4,23624$$

$$4 \quad f = 4,23540$$

$$3 \quad f = 4,23477$$

$$2 \quad f = 4,23365$$

$$4,23678$$

$$4,23754$$

$$4,23858$$

$$1 f = 4,23073$$

$$4,24136$$

$$4,6 f = 4,23560$$

$$4,23633$$

$$3,6 f = 4,23501$$

$$4,23691$$

$$2,6 f = 4,23425$$

$$4,23770$$

МЕДБ

ΔЮРА16

СТАНБ

$$M \pm 0,0012 \quad 40,335$$

$$13,242$$

$$36,928$$

$$d \pm 0,05 \text{ мм} \quad 12,00$$

$$12,35$$

$$11,90 \pm 0,05 \text{ мм}$$

$$L \pm 0,05 \text{ мм} \quad 40,55$$

$$41,35$$

$$41,10$$

СТЕРЖЕНЬ МЕДБ

$$L = 527 \pm 1 \text{ мм}$$

$$d = 9,95 \pm 0,05 \text{ мм}$$

$$f_1 = 3,80515$$

$$f_2 = 7,61069$$

$$f_3 = 11,4128$$

$$f_4 = 15,2145$$

$$f_5 = 19,0130$$